

The expression $\tanh(wx + b)$ represents the hyperbolic tangent function applied to a linear combination of a variable x , a weight w , and a bias b . Here's a breakdown of the components:

1. $w x + b$: This is a linear transformation of the input x , where:
 - w is a weight (scalar or matrix, depending on the context).
 - b is a bias term (scalar or vector).
 - x is the input variable (scalar or vector).
2. $\tanh(\cdot)$: The hyperbolic tangent function is defined as:

$$\tanh(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}},$$

where $z = wx + b$. The output of $\tanh(z)$ is a value between -1 and 1 .

Key Properties of $\tanh(wx + b)$:

- **Range**: The output is bounded between -1 and 1 .
- **S-Shaped Curve**: The function is smooth and differentiable, making it useful in machine learning (e.g., as an activation function in neural networks).
- **Centered at Zero**: Unlike the sigmoid function, $\tanh$ is symmetric around zero, which can help with convergence in training models.

Example:

If $w = 2$, $b = 1$, and $x = 3$, then:

$$\tanh(wx + b) = \tanh(2 \cdot 3 + 1) = \tanh(7).$$

Using the definition of $\tanh$:

$$\tanh(7) = \frac{e^7 - e^{-7}}{e^7 + e^{-7}} \approx 1 \quad (\text{since } e^7 \text{ is much larger than } e^{-7}).$$

Let me know if you'd like further clarification or examples!



draw the S-Shaped Curve for this function



The **S-shaped curve** for the function $\tanh(wx + b)$ is a characteristic plot of the hyperbolic tangent function. Here's how you can visualize it:

Key Features of the Curve:

1. **Shape**: Smooth, S-shaped (sigmoidal) curve.
2. **Range**: Output values are bounded between -1 and 1 .
3. **Symmetry**: The function is symmetric around the origin $(0, 0)$.
4. **Slope**: The slope is steepest around $w x + b = 0$ and flattens out as $w x + b$ approaches $\pm\infty$.

How to Draw the Curve:

1. **Horizontal Axis** ($w x + b$): Represents the input to the $\tanh$ function.
2. **Vertical Axis** ($\tanh(w x + b)$): Represents the output of the $\tanh$ function.

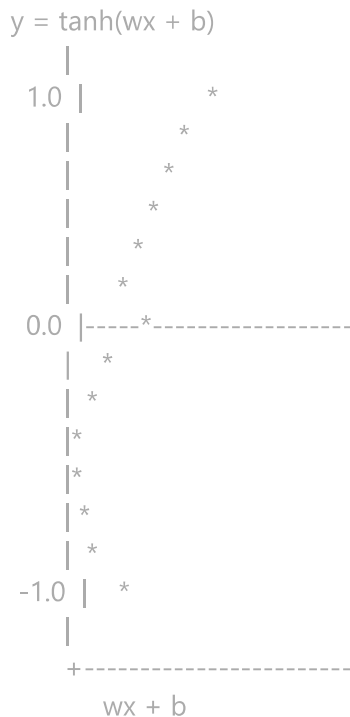
Steps:

1. For large negative values of $w x + b$ (e.g., $w x + b \rightarrow -\infty$):
$$\tanh(w x + b) \approx -1.$$
2. At $w x + b = 0$:
$$\tanh(0) = 0.$$
3. For large positive values of $w x + b$ (e.g., $w x + b \rightarrow +\infty$):
$$\tanh(w x + b) \approx 1.$$

Example Plot:

Here's a textual representation of the curve:

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Python Code to Plot the Curve:

If you'd like to generate the plot programmatically, here's an example using Python and Matplotlib:

python

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```

import numpy as np
import matplotlib.pyplot as plt

# Define the tanh function
def tanh(w, b, x):
    return np.tanh(w * x + b)

# Generate x values
x = np.linspace(-10, 10, 1000) # Range of x values
w = 1 # Weight
b = 0 # Bias
y = tanh(w, b, x) # Compute tanh(wx + b)

# Plot the curve
plt.plot(x, y, label=f"tanh({w}x + {b})")
plt.axhline(0, color="black", linestyle="--", linewidth=0.5) # Horizontal line at y=0
plt.axvline(0, color="black", linestyle="--", linewidth=0.5) # Vertical line at x=0
plt.xlabel("wx + b")
plt.ylabel("tanh(wx + b)")
plt.title("S-Shaped Curve of tanh(wx + b)")
plt.legend()
plt.grid()
plt.show()

```

Output of the Code:

The plot will show the S-shaped curve of $\tanh(wx + b)$, with:

- y approaching -1 as $w x + b \rightarrow -\infty$,
- $y = 0$ at $w x + b = 0$,
- y approaching 1 as $w x + b \rightarrow +\infty$.

Let me know if you need further clarification!